

Introduction to Cloud Security

Cloud security protects cloud-based **data, applications and infrastructure** from cyber threats. The top cloud providers – AWS, Azure and GCP offer security tools to mitigate risks.

Basics of Cloud Security (AWS, Azure, GCP)

AWS Security

1. IAM (Identity & Access Management) → Controls user access
2. AWS Shield → DDoS protection
3. AWS WAF → Blocks web attacks (SQL Injection, XSS)
4. GuardDuty → AI powered threat detection
5. CloudTrail → Tracks API activity

Azure Security

1. Azure AD → Manages authentication & MFA
2. Security Center → Monitors security risks.
3. Sentinel → AI based threat detection (SIEM)
4. Azure Firewall → Protects workloads
5. Key Vault → Secures cryptographic keys

GCP Security

1. Security Command Center → Detects threat & misconfigurations
2. IAM → Controls user roles & permissions
3. Cloud Armor → Ddos protection
4. BeyondCorp → Zero Trust security model
5. Secret Manager → Stores API Keys securely

Security Feature	AWS	Azure	GCP
Access Control	IAM	Azure AD	IAM
Threat Detection	GuardDuty	Security Center	Security Command Center
DDoS Protection	AWS Shield	Azure DDoS Protection	Cloud Armor
Firewall	AWS WAF	Azure Firewall	VPC Firewall Rules
Security Monitoring	CloudTrail	Sentinel (SIEM)	Security Command Center
Data Encryption	AWS KMS	Azure Key Vault	Secret Manager
Zero Trust Security	AWS Zero Trust Framework	Azure Zero Trust Model	BeyondCorp
Compliance	AWS Artifact	Azure Compliance Manager	Google Cloud Compliance

Shared Responsibility Model

- defines **security responsibilities** between cloud provider (AWS, Azure, GCP) and customers(users/organization)

Who is Responsible for What?		
Responsibility	Cloud Provider (AWS, Azure, GCP)	Customer (You/Organization)
Cloud Infrastructure Security (Physical security, networking, hardware)	✓ Yes	× No
Operating System Security (Managed Services)	✓ Yes	× No
Data Security & Encryption	⚠ Shared	✓ Yes
Application Security (Code, updates, patches)	× No	✓ Yes
User Access Control (IAM, Permissions, MFA)	× No	✓ Yes
Network Configuration & Firewall Rules	⚠ Shared	✓ Yes
Compliance & Legal Requirements	✓ Yes (for cloud infrastructure)	✓ Yes (for data & applications)

Why?

- prevents misunderstanding about security responsibilities.
- Helps organizations configure cloud security properly.
- Minimizes risk of misconfigurations & data breaches.

Common cloud security threats

Cloud Security threats can **expose sensitive data, disrupt operations, and lead to financial losses**. Here are three major threats with real incidents;

1. Misconfigurations

- Incorrect cloud settings can expose sensitive data
eg : public access to storage, weak IAM policies)

Case :

Capital One Data Breach (2019) :

- A misconfigured AWS S3 bucket exposed 100 million customers records, including SSNs and credit card details.
- The attackers exploited overly permissive IAM roles to access sensitive files.
- Capital One faced \$80 million in fines and severe reputation damage.

2. Insecure APIs

- API : allow cloud apps to communicate.
If not secured, attackers can steal data, manipulate requests, or gain unauthorized access.

Case :

T-Mobile API Breach(2018) :

- Hackers exploited an insecure API to access 2 million customer records.
- API lacked proper authentication and security checks.
- Resulted in data leaks, fraud risks and customer distrust.

3. Data Breaches & Unauthorized Access

- Weak access controls, stolen credentials, or poor encryption can lead to data leaks.
- Cloud environments often store sensitive corporate/personal information.

Case :

Facebook Data Leak (2021) :

- Over 533 million Facebook users records were leaked due to an unprotected cloud database .
- Data included phone numbers, emails, and personal info.
- Attackers scraped the data and sold it on hacking

Misconfigurations → Can expose millions of records.
Insecure APIs → Can lead to unauthorized access & fraud.
Data Breaches → Can cause financial & legal consequences.

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~week 5 task 1